INTRODUCTION

EGOR GRISHIN
STUDENT ID: 125380990018

№1

In a general supervised machine learning problem we try to find a function $g(x)$ to predict Y based on a set of variables X. One of the predictors is the conditional expectation:

$$f^*(x) = E(Y \mid X)$$

Prove that it is the best predictor based on the squared loss function. That is, for any $g(x)$, we have

$$E[(Y - f^*(X))^2] \leq E[(Y - g(X))^2]$$

We want to show that for any measurable function $g(X)$,

$$\mathbb{E}\big[(Y - f^*(X))^2\big] \leq \mathbb{E}\big[(Y - g(X))^2\big],$$

where

$$f^*(X) = \mathbb{E}[Y|X].$$

We can rewrite:

$$Y - g(X) = (Y - f^*(X)) + (f^*(X) - g(X)).$$

Then, by expanding the square:

$$(Y - g(X))^2 = (Y - f^*(X))^2 + 2(Y - f^*(X))(f^*(X) - g(X)) + (f^*(X) - g(X))^2.$$

Taking expectations and using linearity:

$$\mathbb{E}\big[(Y - g(X))^2\big] = \mathbb{E}\big[(Y - f^*(X))^2\big] + 2\,\mathbb{E}[(Y - f^*(X))(f^*(X) - g(X))] + \mathbb{E}\big[(f^*(X) - g(X))^2\big]\,.$$

We have cross-term:

$$\mathbb{E}[(Y - f^*(X))(f^*(X) - g(X))] = \mathbb{E}[\mathbb{E}[(Y - f^*(X))(f^*(X) - g(X)) \mid X]]\,.$$

Since $(f^*(X) - g(X))$ depends only on X, it can be taken out of the inner conditional expectation:

$$= \mathbb{E}[(f^*(X) - g(X))\,\mathbb{E}[Y - f^*(X) \mid X]]\,.$$

But by the definition of conditional expectation,

$$\mathbb{E}[Y - f^*(X) \mid X] = \mathbb{E}[Y \mid X] - f^*(X) = 0.$$

Hence,

$$\begin{aligned}\mathbb{E}\big[(Y - f^*(X))(f^*(X) - g(X))\big] &= \mathbb{E}\Big[\mathbb{E}\big[(Y - f^*(X))(f^*(X) - g(X)) \mid X\big]\Big] \\ &= \mathbb{E}\Big[(f^*(X) - g(X))\,\mathbb{E}[Y - f^*(X) \mid X]\Big] \\ &= \mathbb{E}\big[(f^*(X) - g(X)) \cdot 0\big] = 0.\end{aligned}$$

Therefore,

$$\mathbb{E}\big[(Y - g(X))^2\big] = \mathbb{E}\big[(Y - f^*(X))^2\big] + \mathbb{E}\big[(f^*(X) - g(X))^2\big].$$

Since $\mathbb{E}\big[(f^*(X) - g(X))^2\big] \geq 0$, we get

$$\mathbb{E}\big[(Y - g(X))^2\big] \geq \mathbb{E}\big[(Y - f^*(X))^2\big].$$

Equality holds **if and only if**

$$\mathbb{E}\big[(f^*(X) - g(X))^2\big] = 0,$$

№2

Suppose we have the standard Wiener process $W(t)$, the following is called an integrated Wiener process

$$\int_0^t W(s)\,ds.$$

What is the variance of this integrated process?

Let $W(t)$ be a standard Wiener process (Brownian motion) with
$\mathbb{E}[W(t)] = 0, \quad \text{and} \quad \mathrm{Cov}(W(s), W(u)) = \min(s, u)$. Define the **integrated Wiener process**:

$$I_t = \int_0^t W(s)\,ds.$$

Since $\mathbb{E}[I_t] = 0$,

$$\mathrm{Var}(I_t) = \mathbb{E}[I_t^2] = \mathbb{E}\left[\int_0^t \int_0^t W(s)W(u)\,ds\,du\right].$$

By linearity of expectation,

$$\mathrm{Var}(I_t) = \int_0^t \int_0^t \mathbb{E}[W(s)W(u)]\, ds\, du = \int_0^t \int_0^t \min(s,u)\, ds\, du.$$

Split the square domain into two regions ($s < u$ and $s > u$):

$$\int_0^t \int_0^t \min(s,u)\, ds\, du = \int_0^t \left(\int_0^u s\, ds + \int_u^t u\, ds \right) du.$$

Compute each part:

$$\int_0^u s\, ds = \frac{u^2}{2}, \qquad \int_u^t u\, ds = u(t-u).$$

Substitute:

$$\int_0^t \left(\frac{u^2}{2} + u(t-u) \right) du = \int_0^t \left(ut - \frac{u^2}{2} \right) du.$$

Integrate:

$$\left[\frac{u^2 t}{2} - \frac{u^3}{6} \right]_0^t = \frac{t^3}{2} - \frac{t^3}{6} = \frac{t^3}{3}.$$

Hence,

$$\mathrm{Var}\left(\int_0^t W(s)\, ds \right) = \frac{t^3}{3}.$$

Alternative Proof via **Itô Isometry**

Using integration by parts:

$$\int_0^t W(s)\, ds = \int_0^t (t-s)\, dW(s).$$

Then by Itô isometry:

$$\mathrm{Var}(I_t) = \mathbb{E}\left[\left(\int_0^t (t-s)\, dW(s) \right)^2 \right] = \int_0^t (t-s)^2\, ds = \frac{t^3}{3}.$$

№3

We look at a process $X(t)$ over a time interval $[0,t]$ and study how to approximate this stochastic process using a random walk model. We divide the time interval into many small time steps $\Delta t = \frac{t}{n}$. At each time unit we have a symmetric random walk, moving up or down by Δx unit with equal probability $1/2$. That is, Y_i has a value 1 if it moves up and a value of -1 if it moves down.

After n steps, the position of process X_t could be written as follows

$$X(t) = \Delta x(Y_1 + Y_2 + \cdots + Y_{\left[\frac{t}{\Delta t}\right]})$$

where $\left[\frac{t}{\Delta t}\right]$ is the largest integer less than or equal to $\frac{t}{\Delta t}$, and the Y_i's are assumed independent with

$$Pr[Y_i = 1] = Pr[Y_i = -1] = \frac{1}{2}.$$

№3 (a)

Prove

$$E(X(t)) = 0$$
$$Var(X(t)) = (\Delta x)^2 \left\{ \frac{t}{\Delta t} \right\}$$

We consider a symmetric random walk approximation to a stochastic process $X(t)$ on the interval $[0,t]$.
We divide the interval into small steps of size Δt and define $m = \left\lfloor \frac{t}{\Delta t} \right\rfloor$.
At each time step, the process moves **up or down** by Δx with equal probability $1/2$:

$$Y_i = \begin{cases} +1, & \text{with probability } \frac{1}{2}, \\ -1, & \text{with probability } \frac{1}{2}. \end{cases}$$

The position after m steps is:

$$X(t) = \Delta x(Y_1 + Y_2 + \cdots + Y_m) = \Delta x \sum_{i=1}^m Y_i.$$

Each Y_i is independent and identically distributed with:

$$\mathbb{E}[Y_i] = 0, \qquad \mathrm{Var}(Y_i) = 1.$$

Using linearity of expectation:

$$\mathbb{E}[X(t)] = \Delta x \sum_{i=1}^m \mathbb{E}[Y_i] = \Delta x \cdot m \cdot 0 = 0.$$

Thus:

$$\mathbb{E}[X(t)] = 0.$$

Since the Y_i's are independent:

$$\mathrm{Var}(X(t)) = \mathrm{Var}\left(\Delta x \sum_{i=1}^{m} Y_i\right) = (\Delta x)^2 \,\mathrm{Var}\left(\sum_{i=1}^{m} Y_i\right) = (\Delta x)^2 \sum_{i=1}^{m} \mathrm{Var}(Y_i).$$

Each $\mathrm{Var}(Y_i) = 1$, so:

$$\mathrm{Var}(X(t)) = (\Delta x)^2 \, m.$$

Substituting $m = \left\lfloor \frac{t}{\Delta t} \right\rfloor$:

$$\mathrm{Var}(X(t)) = (\Delta x)^2 \left\lfloor \frac{t}{\Delta t} \right\rfloor.$$

№3 (b)

Under which condition that we would have the following? How do you state this conditional in a simple non-math language?

$$E(X(t)) = 0$$
$$Var(X(t)) \to \sigma^2 t$$

We start with the discrete random walk model:

$$X(t) = \Delta x \sum_{i=1}^{m} Y_i, \qquad m = \frac{t}{\Delta t},$$

where each Y_i is independent with
$\mathbb{E}[Y_i] = 0$ and $\mathrm{Var}(Y_i) = 1$.
We already have:

$$\mathbb{E}[X(t)] = 0, \qquad \mathrm{Var}(X(t)) = (\Delta x)^2 \frac{t}{\Delta t}.$$

For the random walk to approximate a continuous process (Brownian motion) with variance rate σ^2, we require that as the time steps get smaller:

$$\mathrm{Var}(X(t)) \to \sigma^2 t.$$

This means that:

$$\lim_{\Delta t \to 0} \frac{(\Delta x)^2}{\Delta t} = \sigma^2.$$

or equivalently,

$$\Delta x = \sigma \sqrt{\Delta t}.$$

Intuitive (non-mathematical) explanation:

> As we make the time steps smaller, we must also make the jump size smaller so that **the average squared movement per unit time remains constant**.

> The step size Δx should shrink proportionally to $\sqrt{\Delta t}$.

This ensures that:

- The process doesn't "explode" (variance stays finite),
- The random walk becomes **continuous** in the limit,
- And the limiting process is a **Brownian motion** with volatility σ.

Under the scaling condition $\Delta x = \sigma \sqrt{\Delta t}$,

$$\mathbb{E}[X(t)] = 0, \qquad \mathrm{Var}(X(t)) \to \sigma^2 t,$$

so the random walk converges in distribution to a **Brownian motion** with variance parameter σ^2.

№4

We have observed n observations $x_1, x_2, \ldots, x_n$ at times $t_1, t_2, \ldots, t_n$ from a Brownian motion with a constant drift μ and a constant variation σ. Find the maximum likelihood estimators for μ and σ.

We observe $x_1, x_2, \ldots, x_n$ at times $t_1, t_2, \ldots, t_n$ from a Brownian motion with drift μ and variance parameter σ^2. That is:

$$X(t) \sim \text{Brownian motion with drift } \mu, \text{ diffusion coefficient } \sigma.$$

Meaning:

$$dX(t) = \mu \, dt + \sigma \, dW(t),$$

where $W(t)$ is a standard Wiener process.

Equivalently:

$$X(t_i) - X(t_{i-1}) \sim \mathcal{N}\big(\mu(t_i - t_{i-1}),\, \sigma^2(t_i - t_{i-1})\big),$$

and increments are independent.

Let $\Delta t_i = t_i - t_{i-1}$ for $i = 1, \ldots, n$, with $t_0 = 0$ and possibly $X(0) = x_0$ known.

Then:

$$X(t_1) \sim \mathcal{N}(x_0 + \mu t_1,\, \sigma^2 t_1).$$

It is better to work with **increments**:

$$y_i = x_i - x_{i-1}, \qquad i = 1, \ldots, n.$$

If x_0 is given, or even if not, we can estimate μ and σ from the increments.

The joint density of $(x_1, \ldots, x_n)$ given x_0 is:

$$f(x_1, \ldots, x_n \mid x_0, \mu, \sigma) = \prod_{i=1}^{n} \frac{1}{\sqrt{2\pi\sigma^2 \Delta t_i}} \exp\left[-\frac{(x_i - x_{i-1} - \mu\Delta t_i)^2}{2\sigma^2 \Delta t_i}\right].$$

Let $\Delta x_i = x_i - x_{i-1}$ and $\Delta t_i = t_i - t_{i-1}$.

Then the log-likelihood is:

$$\ell(\mu, \sigma^2) = -\frac{n}{2}\log(2\pi) - \frac{n}{2}\log(\sigma^2) - \frac{1}{2}\sum_{i=1}^{n}\log(\Delta t_i) - \frac{1}{2\sigma^2}\sum_{i=1}^{n}\frac{(\Delta x_i - \mu\Delta t_i)^2}{\Delta t_i}.$$

Differentiate with respect to μ:

$$\frac{\partial \ell}{\partial \mu} = \frac{1}{\sigma^2}\sum_{i=1}^{n}(\Delta x_i - \mu\Delta t_i) = 0.$$

Solve for μ:

$$\sum_{i=1}^{n}\Delta x_i - \mu\sum_{i=1}^{n}\Delta t_i = 0,$$

hence:

$$\hat{\mu} = \frac{\sum_{i=1}^{n}\Delta x_i}{\sum_{i=1}^{n}\Delta t_i}.$$

But note that:

$$\sum_{i=1}^{n}\Delta x_i = x_n - x_0, \quad \sum_{i=1}^{n}\Delta t_i = t_n - t_0.$$

Therefore:

$$\hat{\mu} = \frac{x_n - x_0}{t_n - t_0}.$$

Differentiate ℓ with respect to σ^2:

Let $\theta = \sigma^2$. Then:

$$\frac{\partial \ell}{\partial \theta} = -\frac{n}{2\theta} + \frac{1}{2\theta^2}\sum_{i=1}^{n}\frac{(\Delta x_i - \mu\Delta t_i)^2}{\Delta t_i}.$$

Set to zero:

$$-\frac{n}{2\theta} + \frac{1}{2\theta^2}\sum_{i=1}^{n}\frac{(\Delta x_i - \mu\Delta t_i)^2}{\Delta t_i} = 0.$$

Simplify:

$$n\theta = \sum_{i=1}^{n}\frac{(\Delta x_i - \mu\Delta t_i)^2}{\Delta t_i}.$$

Hence the MLE is:

$$\hat{\sigma}^2 = \frac{1}{n}\sum_{i=1}^{n}\frac{(\Delta x_i - \hat{\mu}\,\Delta t_i)^2}{\Delta t_i}.$$

$$\hat{\mu} = \frac{x_n - x_0}{t_n - t_0}, \qquad \hat{\sigma}^2 = \frac{1}{n}\sum_{i=1}^{n}\frac{(\Delta x_i - \hat{\mu}\,\Delta t_i)^2}{\Delta t_i}.$$

These are the **maximum likelihood estimators** for the drift μ and the diffusion coefficient σ^2 of a Brownian motion with constant drift and variance, given discrete observations at possibly irregular times.

№5

$$dX(t) = (aX(t) + b)dt + c\,dW(t)$$

Solve this equation. (Hint: you could apply Ito's lemma to $X(t)e^{-at}$.)

We have the stochastic differential equation:

$$dX(t) = (aX(t) + b)\,dt + c\,dW(t), \qquad X(0) = X_0.$$

Multiply both sides by e^{-at} and define

$$Y(t) = e^{-at}X(t).$$

By Itô's lemma (or Itô's product rule):

$$dY(t) = e^{-at}\,dX(t) + X(t)\,d(e^{-at}) + d\langle e^{-at}, X(t)\rangle.$$

Since e^{-at} is deterministic, the quadratic variation term vanishes.

Thus:

$$dY(t) = e^{-at}\big[(aX(t)+b)\,dt + c\,dW(t)\big] - ae^{-at}X(t)\,dt.$$

Simplify:

$$dY(t) = e^{-at}\big[b\,dt + c\,dW(t)\big].$$

Integrate both sides:

$$Y(t) - Y(0) = \int_0^t e^{-as} b\,ds + \int_0^t e^{-as} c\,dW_s.$$

Since $Y(0) = X_0$,

$$Y(t) = X_0 + b\int_0^t e^{-as}\,ds + c\int_0^t e^{-as}\,dW_s.$$

Multiply both sides by e^{at}:

$$X(t) = e^{at}X_0 + be^{at}\int_0^t e^{-as}\,ds + ce^{at}\int_0^t e^{-as}\,dW_s.$$

If $a \neq 0$, then:

$$\int_0^t e^{-as}\,ds = \frac{1-e^{-at}}{a}.$$

Substitute back:

$$X(t) = e^{at}X_0 + \frac{b}{a}(e^{at}-1) + ce^{at}\int_0^t e^{-as}\,dW_s.$$

If $a = 0$, the SDE reduces to:

$$dX(t) = b\,dt + c\,dW(t),$$

so the solution is:

$$X(t) = X_0 + bt + cW_t.$$

Since $\mathbb{E}\left[\int_0^t e^{-as}\,dW_s\right] = 0$, we have:

$$\mathbb{E}[X(t)] = e^{at}X_0 + \frac{b}{a}(e^{at}-1),$$

and

$$\mathrm{Var}[X(t)] = c^2 e^{2at}\int_0^t e^{-2as}\,ds = \frac{c^2}{2a}(e^{2at}-1).$$

№6

For a discrete probability density function $f(x)$ for a random variable which takes values on a set $(x_1, x_2, \ldots, x_n)$, the entropy is defined as

$$H(X) = E[-\ln(f(x))] = -\sum_{i=1}^{n} f(x_i)\ln[f(x_i)]$$

For a continuous random variable X with density function $f(x)$, the entropy is defined as

$$H(X) = E[-\ln(f(x))] = -\int f(x)\ln[f(x)]\,dx$$

№6 (a)

Using the discrete entropy definition with n possible outcomes to show that the uniform distribution has the maximum entropy.

Let

$p_i = \Pr[X = x_i], \quad p_i \geq 0, \quad \sum_{i=1}^{n} p_i = 1.$

Entropy (using natural logs) is defined as

$$H(p) = -\sum_{i=1}^{n} p_i \ln p_i$$

Lagrangian

$$L(p,\lambda) = -\sum_{i=1}^{n} p_i \ln p_i + \lambda\left(\sum_{i=1}^{n} p_i - 1\right)$$

First-Order Conditions (for any $p_j > 0$)

$$\frac{\partial L}{\partial p_j} = -(\ln p_j + 1) + \lambda = 0$$

Hence

$$\ln p_j = \lambda - 1 \quad \Rightarrow \quad p_j = c \quad (\text{same for all } j)$$

Using the constraint $\sum_i p_i = 1$ gives

$$c = \frac{1}{n}$$

Thus, the only stationary point in the simplex's interior is

$$p_1 = p_2 = \cdots = p_n = \frac{1}{n}$$

The entropy over n outcomes is **maximized uniquely** by the **uniform distribution**

$$p_i = \frac{1}{n}, \quad \forall i.$$

№6 (b)

For the normal random variable $X \sim N(\mu, \sigma^2)$,

$$H(X) = \frac{1 + \ln(2\pi\sigma^2)}{2}.$$

For $X \sim \mathcal{N}(\mu, \sigma^2)$, the density is

$$f(x) = \frac{1}{\sqrt{2\pi}\,\sigma} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right).$$

Then

$$\ln f(x) \;=\; -\tfrac{1}{2}\ln(2\pi) \;-\; \ln\sigma \;-\; \frac{(x-\mu)^2}{2\sigma^2}.$$

Entropy:

$$H(X) \;=\; -\int f(x)\,\ln f(x)\,dx \;=\; -\,\mathbb{E}\big[\ln f(X)\big].$$

Substitute $\ln f(X)$:

$$\begin{aligned} H(X) &= -\,\mathbb{E}\left[-\tfrac{1}{2}\ln(2\pi) - \ln\sigma - \frac{(X-\mu)^2}{2\sigma^2}\right] \\ &= \tfrac{1}{2}\ln(2\pi) + \ln\sigma + \frac{1}{2\sigma^2}\,\mathbb{E}\big[(X-\mu)^2\big]. \end{aligned}$$

But $\mathbb{E}\big[(X-\mu)^2\big] = \sigma^2$. Hence

$$H(X) \;=\; \tfrac{1}{2}\ln(2\pi) + \ln\sigma + \tfrac{1}{2} \;=\; \tfrac{1}{2}\Big(1 + \ln(2\pi\sigma^2)\Big).$$

№6 (c)

Suppose that a discrete random variable has three possible outcomes, and whose mean is 2.5, what is the maximum entropy distribution with this mean constraint?

We want to maximize the entropy

$$H(p) = -\sum_{i=1}^{3} p_i \ln p_i$$

subject to the constraints

$$p_1 + p_2 + p_3 = 1, \qquad \mathbb{E}[X] = 1 \cdot p_1 + 2 \cdot p_2 + 3 \cdot p_3 = 2.5, \qquad p_i \geq 0.$$

Introduce multipliers μ, λ and form

$$L = -\sum_{i=1}^{3} p_i \ln p_i + \mu\left(\sum_{i=1}^{3} p_i - 1\right) + \lambda\left(\sum_{i=1}^{3} i\,p_i - 2.5\right).$$

First-Order Conditions (for each i with $p_i > 0$)

$$\frac{\partial L}{\partial p_i} = -(\ln p_i + 1) + \mu + \lambda i = 0 \quad \Longrightarrow \quad \ln p_i = \mu - 1 + \lambda i \quad \Longrightarrow \quad p_i \propto e^{\lambda i}.$$

Thus the maximizing distribution has exponential form

$$p_i = \frac{e^{\lambda i}}{Z}, \qquad Z = e^{\lambda} + e^{2\lambda} + e^{3\lambda}.$$

The given mean 2.5 implies

$$\frac{e^{\lambda} + 2e^{2\lambda} + 3e^{3\lambda}}{e^{\lambda} + e^{2\lambda} + e^{3\lambda}} = 2.5.$$

Let $u = e^{\lambda} > 0$. Then

$$\frac{u + 2u^2 + 3u^3}{u + u^2 + u^3} = 2.5 \quad \Longleftrightarrow \quad u^2 - u - 3 = 0 \quad \Longrightarrow \quad u = \frac{1 + \sqrt{13}}{2}.$$

Divide numerator and denominator by u:

$$p_1 = \frac{1}{1+u+u^2}, \qquad p_2 = \frac{u}{1+u+u^2}, \qquad p_3 = \frac{u^2}{1+u+u^2}.$$

Using $u = \frac{1+\sqrt{13}}{2}$, we find

$$1+u+u^2 = 5+\sqrt{13}.$$

Final Closed-Form Probabilities

$$p_1 = \frac{1}{5+\sqrt{13}}, \qquad p_2 = \frac{\frac{1+\sqrt{13}}{2}}{5+\sqrt{13}}, \qquad p_3 = \frac{\left(\frac{1+\sqrt{13}}{2}\right)^2}{5+\sqrt{13}}.$$

Using $\sqrt{13} \approx 3.605551$:

$$p_1 \approx 0.116204, \quad p_2 \approx 0.267592, \quad p_3 \approx 0.616204.$$

These satisfy $p_1 + p_2 + p_3 = 1$ and
$\mathbb{E}[X] = 1 \cdot p_1 + 2 \cdot p_2 + 3 \cdot p_3 = 2.5.$

№7

The ridge regression model could be taken as the ordinary regression model

$$Y = X\beta + \varepsilon$$

with a constraint on the regression coefficient such that the sum of the squared coefficients is minimized. That is, the ridge loss function is to minimize the following:

$$L(\beta) = \|Y - X\beta\|^2 + \lambda\|\beta\|^2$$

where Y is an $n \times 1$ matrix, X is an $n \times p$ matrix, and ε is $n \times 1$ representing the errors, which are independent and have a constant variance σ^2 for all errors, $\varepsilon_1, \varepsilon_2, \ldots, \varepsilon_n$.

№7 (a)

Prove that the ridge estimator

$$\hat{\beta}_{ridge} = (X^T X + \lambda I_{pp})^{-1} X^T Y$$

is as shown above, and show that it is the solution to minimize the ridge loss function.

Let

$$L(\beta) = \|Y - X\beta\|^2 + \lambda\|\beta\|^2 = (Y - X\beta)^\top (Y - X\beta) + \lambda\,\beta^\top\beta.$$

Expand the quadratic:

$$\begin{aligned} L(\beta) &= Y^\top Y - 2\beta^\top X^\top Y + \beta^\top X^\top X\beta + \lambda\,\beta^\top\beta \\ &= Y^\top Y - 2\beta^\top X^\top Y + \beta^\top\big(X^\top X + \lambda I_{pp}\big)\beta. \end{aligned}$$

Take the gradient and set to zero (first-order condition):

$$\nabla_\beta L(\beta) = -2X^\top Y + 2\big(X^\top X + \lambda I_{pp}\big)\beta = 0 \implies \big(X^\top X + \lambda I_{pp}\big)\beta = X^\top Y.$$

For $\lambda > 0$, $X^\top X + \lambda I_{pp}$ is symmetric positive definite and hence invertible.
Therefore,

$$\hat{\beta}_{\text{ridge}} = \big(X^\top X + \lambda I_{pp}\big)^{-1} X^\top Y.$$

№7 (b)

Obtain the expected value of the ridge estimator, and show that it is not an unbiased estimator.

Model:

$$Y = X\beta + \varepsilon, \qquad \mathbb{E}[\varepsilon \mid X] = 0, \qquad \mathrm{Var}(\varepsilon \mid X) = \sigma^2 I_n.$$

Ridge estimator:

$$\hat{\beta}_{\text{ridge}} = (X^\top X + \lambda I_p)^{-1} X^\top Y, \qquad \lambda > 0.$$

Plug the model into the estimator:

$$\begin{aligned} \hat{\beta}_{\text{ridge}} &= (X^\top X + \lambda I)^{-1} X^\top (X\beta + \varepsilon) \\ &= (X^\top X + \lambda I)^{-1} X^\top X\beta \;+\; (X^\top X + \lambda I)^{-1} X^\top \varepsilon. \end{aligned}$$

Take the conditional expectation
Using $\mathbb{E}[\varepsilon \mid X] = 0$,

$$\mathbb{E}[\hat{\beta}_{\text{ridge}} \mid X] = (X^\top X + \lambda I)^{-1} X^\top X\beta =: S_\lambda\,\beta,$$

where

$$S_\lambda \equiv (X^\top X + \lambda I)^{-1} X^\top X.$$

we can rewrite S_λ as

$$S_\lambda = I - \lambda(X^\top X + \lambda I)^{-1}.$$

Hence the **bias** of the ridge estimator conditional on X is

$$\begin{aligned}\mathrm{Bias}(\hat{\beta}_{\text{ridge}} \mid X) &= \mathbb{E}[\hat{\beta}_{\text{ridge}} \mid X] - \beta \\ &= (S_\lambda - I)\beta \\ &= -\lambda(X^\top X + \lambda I)^{-1}\beta.\end{aligned}$$

For $\lambda > 0$, this bias equals zero only in trivial cases such as $\beta = 0$.
Therefore, **ridge regression is not an unbiased estimator** of β.

№7 (c)

Obtain the variance of the ridge estimator.

Model:

$$Y = X\beta + \varepsilon, \qquad \mathbb{E}[\varepsilon \mid X] = 0, \qquad \mathrm{Var}(\varepsilon \mid X) = \sigma^2 I_n.$$

Ridge estimator:

$$\hat{\beta}_{\text{ridge}} = (X^\top X + \lambda I)^{-1} X^\top Y.$$

Covariance (given X):

$$\begin{aligned}\mathrm{Var}(\hat{\beta}_{\text{ridge}} \mid X) &= \mathrm{Var}\big((X^\top X + \lambda I)^{-1} X^\top Y \mid X\big) \\ &= (X^\top X + \lambda I)^{-1} X^\top \mathrm{Var}(Y \mid X)\, X\, (X^\top X + \lambda I)^{-1} \\ &= \sigma^2 (X^\top X + \lambda I)^{-1} X^\top X (X^\top X + \lambda I)^{-1}.\end{aligned}$$

Since $X^\top X + \lambda I$ is symmetric positive definite for $\lambda > 0$, its inverse is also symmetric.

№7 (d)

Show that the variance of the ridge estimator is less than the variance of the maximum likelihood estimator for the ordinary regression model.

OLS / MLE:

$$\mathrm{Var}(\hat{\beta}_{\text{OLS}} \mid X) = \sigma^2 A^{-1},$$

where $A = X^\top X$.
Ridge (with $\lambda > 0$):

$$\mathrm{Var}(\hat{\beta}_{\text{ridge}} \mid X) = \sigma^2 (A + \lambda I)^{-1} A (A + \lambda I)^{-1}.$$

Let $A = QDQ^\top$ with Q orthogonal and
$D = \mathrm{diag}(d_1, \ldots, d_p)$, where $d_i > 0$.
Then

$$\mathrm{Var}(\hat{\beta}_{\text{OLS}} \mid X) = \sigma^2 Q\, \mathrm{diag}\left(\frac{1}{d_i}\right) Q^\top,$$

and

$$\mathrm{Var}(\hat{\beta}_{\text{ridge}} \mid X) = \sigma^2 Q\, \mathrm{diag}\left(\frac{d_i}{(d_i + \lambda)^2}\right) Q^\top.$$

For each eigenvalue $d_i > 0$ and $\lambda > 0$:

$$\frac{d_i}{(d_i + \lambda)^2} \le \frac{1}{d_i} \iff d_i^2 \le (d_i + \lambda)^2 \iff 0 \le 2\lambda d_i + \lambda^2,$$

which holds (strictly if $\lambda > 0$).
Therefore, entrywise in the eigenbasis:

$$\mathrm{diag}\left(\frac{d_i}{(d_i + \lambda)^2}\right) \preceq \mathrm{diag}\left(\frac{1}{d_i}\right),$$

and by congruence with Q,

$$\mathrm{Var}(\hat{\beta}_{\text{ridge}} \mid X) \preceq \mathrm{Var}(\hat{\beta}_{\text{OLS}} \mid X).$$

That is, **ridge regression has smaller variance** than OLS, componentwise in the eigenbasis of $X^\top X$.

№7 (e)

From the perspective of the mean squared error (MSE), could the ridge estimator have a smaller MSE than the MLE for the ordinary regression model?

From the **Mean Squared Error (MSE)** point of view, the ridge estimator can have a smaller MSE than the OLS (or MLE) estimator.

The MSE of an estimator combines variance and bias:

$$\mathrm{MSE}(\hat{\beta}) = \mathrm{Var}(\hat{\beta}) + [\mathrm{Bias}(\hat{\beta})]^2.$$

OLS is **unbiased**, so:

$$\mathrm{MSE}_{\mathrm{OLS}} = \mathrm{Var}(\hat{\beta}_{\mathrm{OLS}}) = \sigma^2 (X^\top X)^{-1}.$$

Ridge adds **bias** but reduces **variance**:

$$\mathrm{Var}(\hat{\beta}_{\mathrm{ridge}}) = \sigma^2 (X^\top X + \lambda I)^{-1} X^\top X (X^\top X + \lambda I)^{-1},$$
$$\mathrm{Bias}(\hat{\beta}_{\mathrm{ridge}}) = -\lambda (X^\top X + \lambda I)^{-1} \beta.$$

When predictors are **highly correlated** or the true coefficients β are **small**,
the **variance reduction** dominates the **bias increase**, leading to:

$$\mathrm{MSE}_{\mathrm{ridge}} < \mathrm{MSE}_{\mathrm{OLS}}.$$

№7 (f)

For the above regression model, we assume that β and σ^2 are not constant, but random variables and follow the distributions as follows: σ^2 follows an inverse Gamma distribution with parameters α_0 and β_0, and $\beta|\sigma^2$ follows a normal distribution with mean 0_p and variance $\sigma^2 I_{pp}/\lambda$, where 0_p is a p-dimensional zero vector, and I_{pp} is a $p \times p$ identity matrix and λ is a constant. Obtain the posterior distribution $p(\beta, \sigma^2 | X, Y)$.

Model

$$Y \mid \beta, \sigma^2 \sim \mathcal{N}(X\beta,\ \sigma^2 I_n), \qquad \mathbb{E}[\varepsilon \mid X] = 0, \quad \mathrm{Var}(\varepsilon \mid X) = \sigma^2 I_n.$$

Prior

$$\beta \mid \sigma^2 \sim \mathcal{N}\left(0,\ \frac{\sigma^2}{\lambda} I_p\right) \quad (\text{precision } \tfrac{\lambda}{\sigma^2} I_p), \qquad \sigma^2 \sim \text{Inv-Gamma}(\alpha_0, \beta_0).$$

Define

$$V_n \ = \ (X^\top X + \lambda I_p)^{-1}, \qquad \mu_n \ = \ V_n X^\top Y, \qquad \alpha_n \ = \ \alpha_0 + \frac{n}{2},$$
$$\beta_n \ = \ \beta_0 + \tfrac{1}{2}\big(Y^\top Y - \mu_n^\top V_n^{-1} \mu_n\big) \ = \ \beta_0 + \tfrac{1}{2} Y^\top \big(I_n - X V_n X^\top\big) Y.$$

Posterior (joint)

$$(\beta, \sigma^2) \mid X, Y \ \sim \ \text{Normal-Inverse-Gamma}\big(\mu_n,\ V_n,\ \alpha_n,\ \beta_n\big).$$

Posterior conditionals

$$\beta \mid \sigma^2, X, Y \sim \mathcal{N}(\mu_n,\ \sigma^2 V_n), \qquad \sigma^2 \mid X, Y \sim \text{Inv-Gamma}(\alpha_n,\ \beta_n).$$

Priors $\sigma^2 \sim \text{Inv-Gamma}(\alpha_0, \beta_0)$
$\beta \mid \sigma^2 \sim \mathcal{N}\left(0, \frac{\sigma^2}{\lambda} I_p\right)$
λ is fixed.
Likelihood

$$Y \mid \beta, \sigma^2 \sim \mathcal{N}(X\beta,\ \sigma^2 I_n)$$

Posterior

$$p(\beta, \sigma^2 \mid X, Y) \propto (\sigma^2)^{-\frac{n+p}{2} - \alpha_0 - 1} \exp\left[-\frac{1}{2\sigma^2}\left(\|Y - X\beta\|^2 + \lambda\|\beta\|^2 + 2\beta_0\right)\right]$$

Conditional Posterior of β

$$\beta \mid \sigma^2, X, Y \sim \mathcal{N}(\mu_{\mathrm{post}},\ \Sigma_{\mathrm{post}})$$

where
$\mu_{\mathrm{post}} = (X^\top X + \lambda I)^{-1} X^\top Y = \hat{\beta}_{\mathrm{ridge}},$
$\Sigma_{\mathrm{post}} = \sigma^2 (X^\top X + \lambda I)^{-1}$ Posterior of σ^2
Let $\mathrm{SSR}(\hat{\beta}_{\mathrm{ridge}}) = \|Y - X\hat{\beta}_{\mathrm{ridge}}\|^2$

$$\alpha_n = \alpha_0 + \frac{n+p}{2}, \qquad \beta_n = \beta_0 + \frac{1}{2}\left(\mathrm{SSR}(\hat{\beta}_{\mathrm{ridge}}) + \lambda\|\hat{\beta}_{\mathrm{ridge}}\|^2\right)$$

Then

$$\sigma^2 \mid X, Y \sim \text{Inv-Gamma}(\alpha_n, \beta_n)$$

Joint Posterior

$$\beta \mid \sigma^2, X, Y \sim \mathcal{N}\left((X^\top X + \lambda I)^{-1} X^\top Y,\ \sigma^2 (X^\top X + \lambda I)^{-1}\right),$$
$$\sigma^2 \mid X, Y \sim \text{Inv-Gamma}\left(\alpha_0 + \frac{n+p}{2},\ \beta_0 + \frac{1}{2}\left(\|Y - X\hat{\beta}_{\mathrm{ridge}}\|^2 + \lambda\|\hat{\beta}_{\mathrm{ridge}}\|^2\right)\right)$$

№7 (g)

Find the posterior distribution $p(\beta|\sigma^2, X, Y)$ and state the linkage with the ridge regression estimator of β.

$$\beta \mid \sigma^2, X, Y \sim \mathcal{N}\left(\hat{\beta}_{\text{ridge}},\ \sigma^2 (X^\top X + \lambda I)^{-1}\right),$$

where

$$\hat{\beta}_{\text{ridge}} = (X^\top X + \lambda I)^{-1} X^\top Y.$$

Link to ridge

- The posterior mean $\mathbb{E}[\beta \mid \sigma^2, X, Y] = \hat{\beta}_{\text{ridge}}$ equals the ridge estimator.
- The Gaussian prior $\beta \mid \sigma^2 \sim \mathcal{N}\left(0, \frac{\sigma^2}{\lambda} I\right)$ induces shrinkage toward zero, which is the Bayesian interpretation of ridge regularization.

№8

What is the general formulation of LASSO regression models? What are the differences with ridge regression? What are those situations where ridge and LASSO should be applied?

The **LASSO (Least Absolute Shrinkage and Selection Operator)** regression adds an **L1 penalty** to the standard least squares objective. For predictors $X = [x_1, x_2, \ldots, x_p]$ and response y, the LASSO estimator $\hat{\beta}$ is defined as:

$$\hat{\beta}_{\text{lasso}} = \arg\min_{\beta_0, \beta} \left\{ \sum_{i=1}^{n} \left(y_i - \beta_0 - \sum_{j=1}^{p} x_{ij} \beta_j \right)^2 + \lambda \sum_{j=1}^{p} |\beta_j| \right\}.$$

where:

- $\lambda \geq 0$ — regularization parameter controlling shrinkage strength
- The L1-norm penalty $\sum_{j=1}^{p} |\beta_j|$ encourages **sparsity** — some coefficients become exactly zero.

The **Ridge regression** estimator adds an **L2 penalty** instead of L1:

$$\hat{\beta}_{\text{ridge}} = \arg\min_{\beta_0, \beta} \left\{ \sum_{i=1}^{n} \left(y_i - \beta_0 - \sum_{j=1}^{p} x_{ij} \beta_j \right)^2 + \lambda \sum_{j=1}^{p} \beta_j^2 \right\}.$$

Key Differences

Feature	LASSO (L1)	Ridge (L2)
Penalty term	$\lambda \sum_j \lvert\beta_j\rvert$	$\lambda \sum_j \beta_j^2$
Shrinkage effect	Shrinks some coefficients **exactly to zero**	Shrinks coefficients **continuously toward zero**, but never exactly zero
Variable selection	Performs automatic feature selection	Keeps all variables
Computation	Iterative (coordinate descent, LARS)	Closed form $(X^T X + \lambda I)^{-1} X^T y$

We use LASSO When:

- Many predictors are irrelevant or redundant.
- Goal includes **variable selection** or **model interpretability**.
- $p > n$ (more features than samples).
- Example: Selecting key genes in bioinformatics or key factors in finance.

We use Ridge When:

- Predictors are **highly correlated (multicollinearity)**.
- All predictors are expected to contribute somewhat.
- You need **stability** and **smooth shrinkage**, not sparsity.
- Example: Macroeconomic forecasting or continuous risk models.